

Background

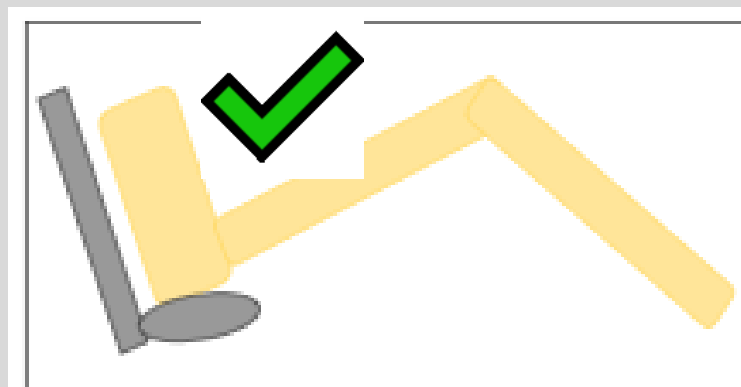
In the past, formula hybrid drivers have reported that ergonomic factors directed towards driver comfort were overlooked in favor of performance. This lead to heavy amounts of lower body fatigue when it came to operating the vehicle during long periods of testing, and especially during the biggest event at competitions: Endurance Races.

Project Goal

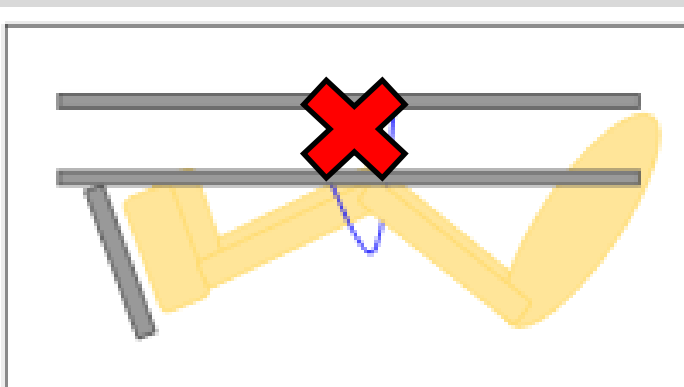
We have created a driver seating system that is built to accommodate multiple sized drivers, comfortably, with the goal of reducing lower-body fatigue while operating the vehicle.

- Two main features
 - Angled Seating System
 - Adjustable Length Pedal System

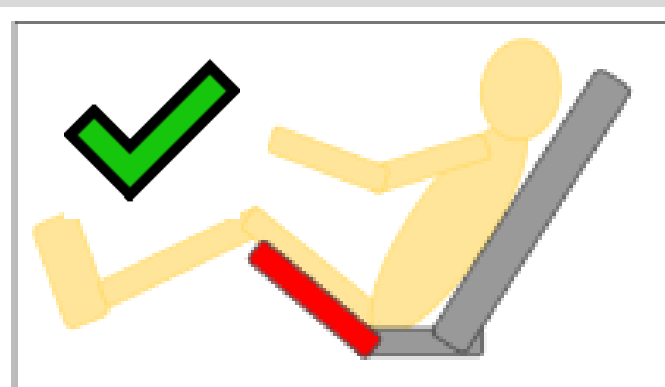
Possible Solutions



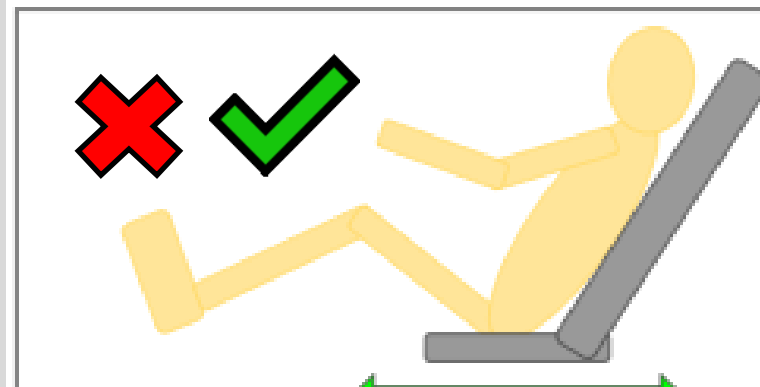
- 1) Footrest for Pedal Box
-Can be added to existing design
-Directly attached, or a standalone part
-May be most beneficial solution given size and theorized effectiveness



- 2) Sling to reduce strain of keeping legs elevated
-Removes all weight from legs
-Easily mounted to frame with hooks/fasteners



- 3) Extended seat to support upper legs
-Likely will provide most comfort
-Most expensive



- 4) Adjustable seat to tune lower body angles for each driver
-Removes need for multiple seats
-Helps every driver reach pedals comfortably
-Can help get driver's head into proper position for 200 degree view

Design Decision Matrix

Ideas	Egress Time	Safety	Comfort	Material	Size	Weight <5lbs	Totals
Foot Rest	5	2	5	3	4	4	23
Sliding Butt Plate	4	3	5	3	3	4	22
Extended Seat	4	3	5	5	4	2	23
Hanging Sling	4	3	4	3	2	3	19

The best designs include Footrest, Extended Seat & Sliding Butt Plate

Materials and Methods

Cockpit Dimensions:

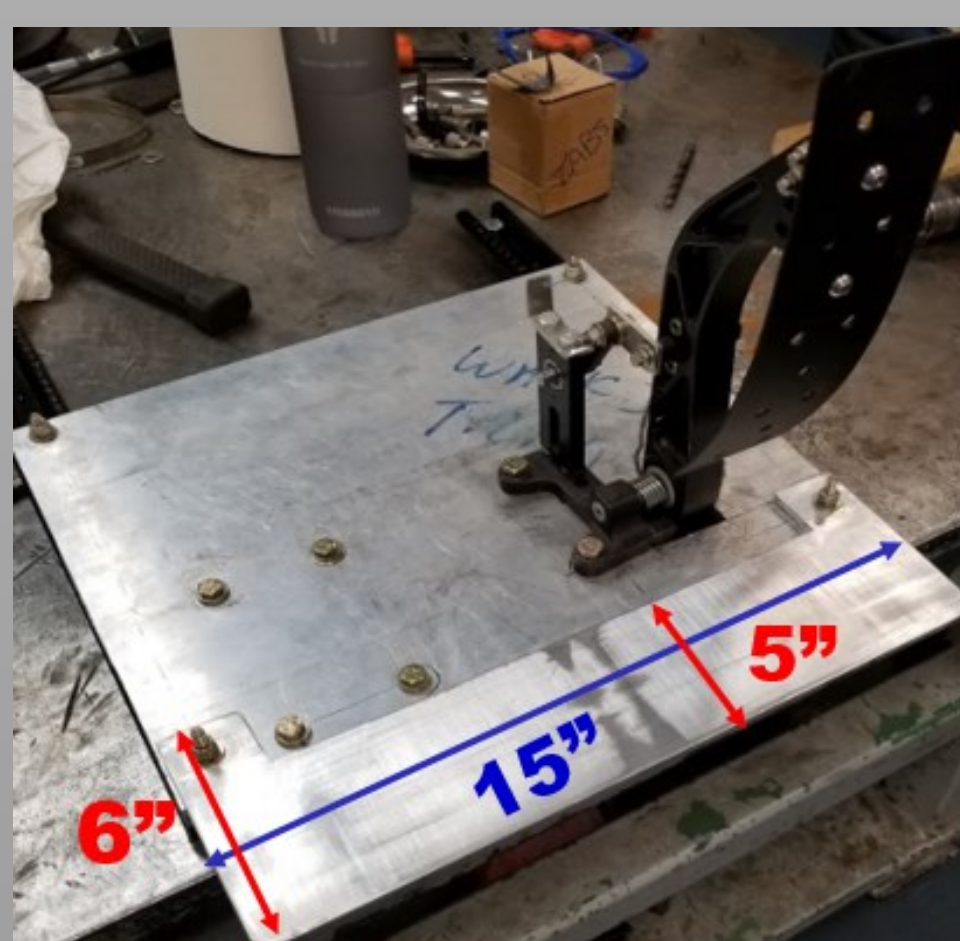
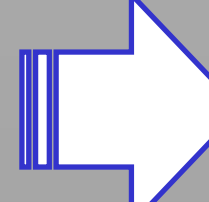
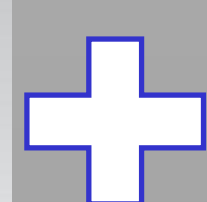
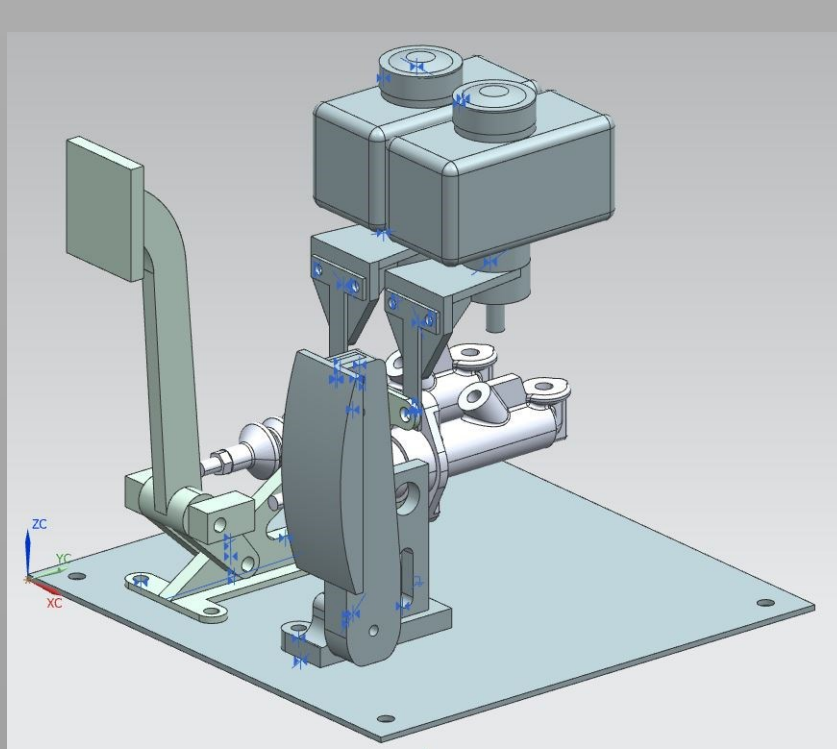
- Seat Back
 - 29" x 21"
- Seat Bottom
 - 16" x 16"
- Seat to Pedal Face
 - 34" ← to → 42"



Vehicle Front ← → Vehicle Rear



Lightweight foam beads, and resin, were combined under vacuum to mold the seat according to specified cockpit dimensions

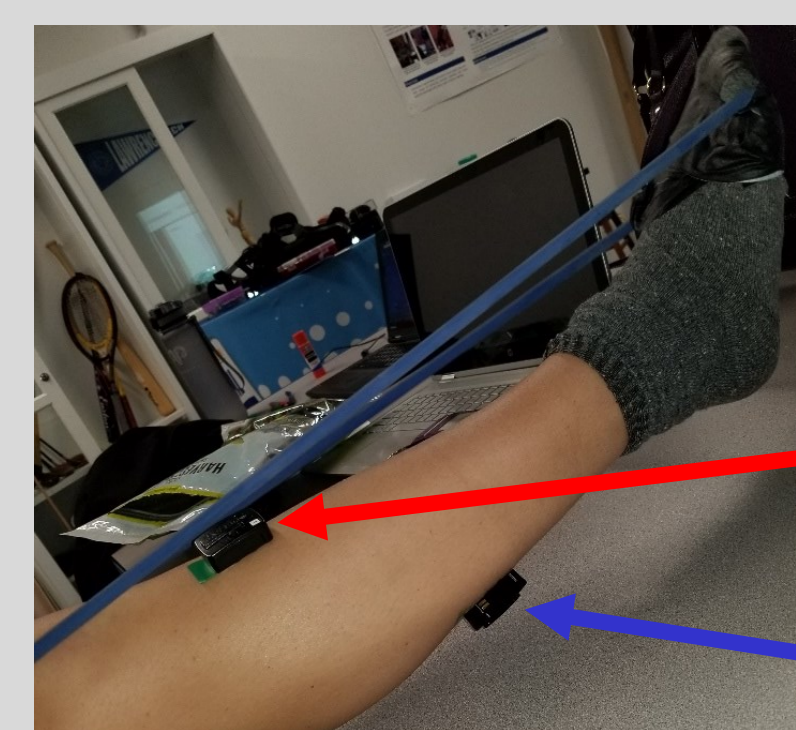
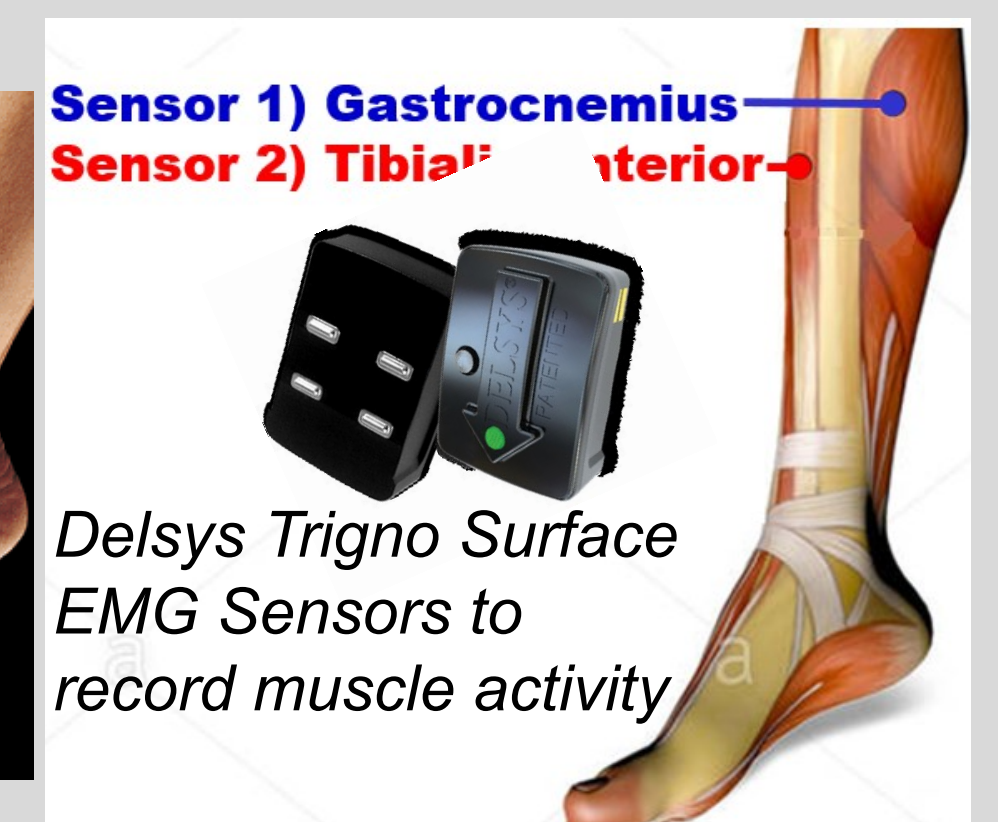
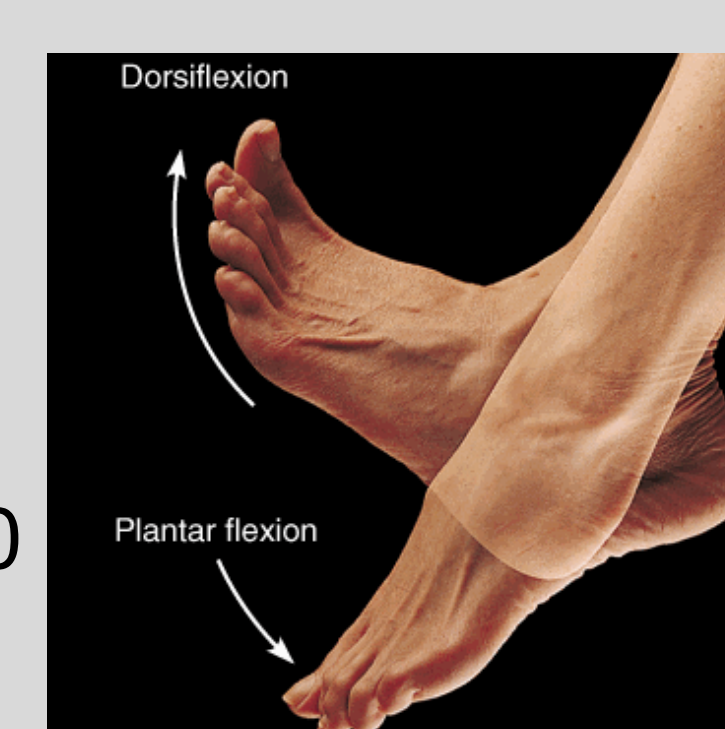


Final Designs

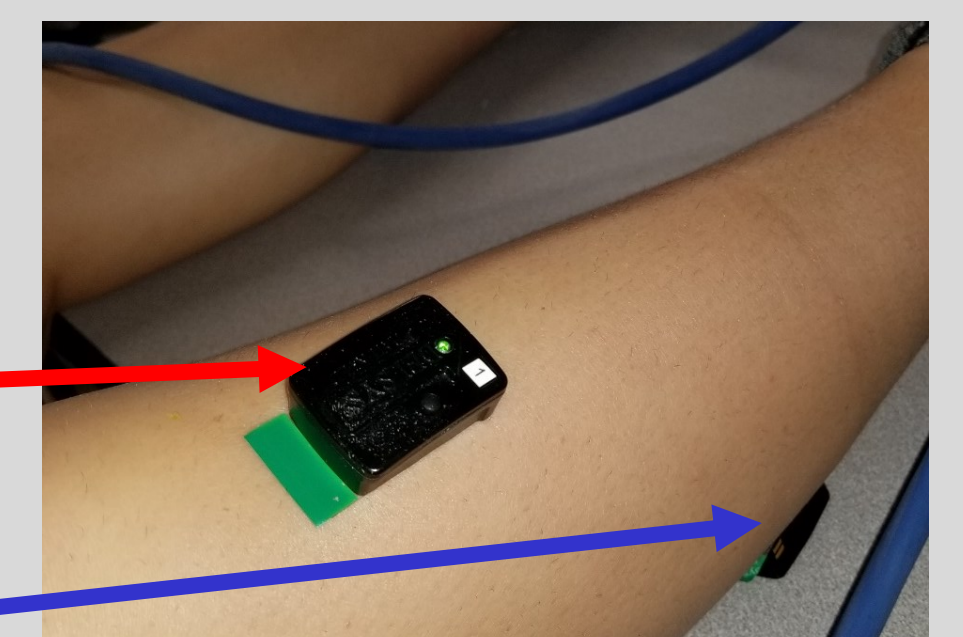


Testing and Results

Test Procedure: Perform pedal activation cycles (Plantar Flexion) until fatigue is felt, then complete 10 more cycles to be recorded and analyzed under fatigue conditions



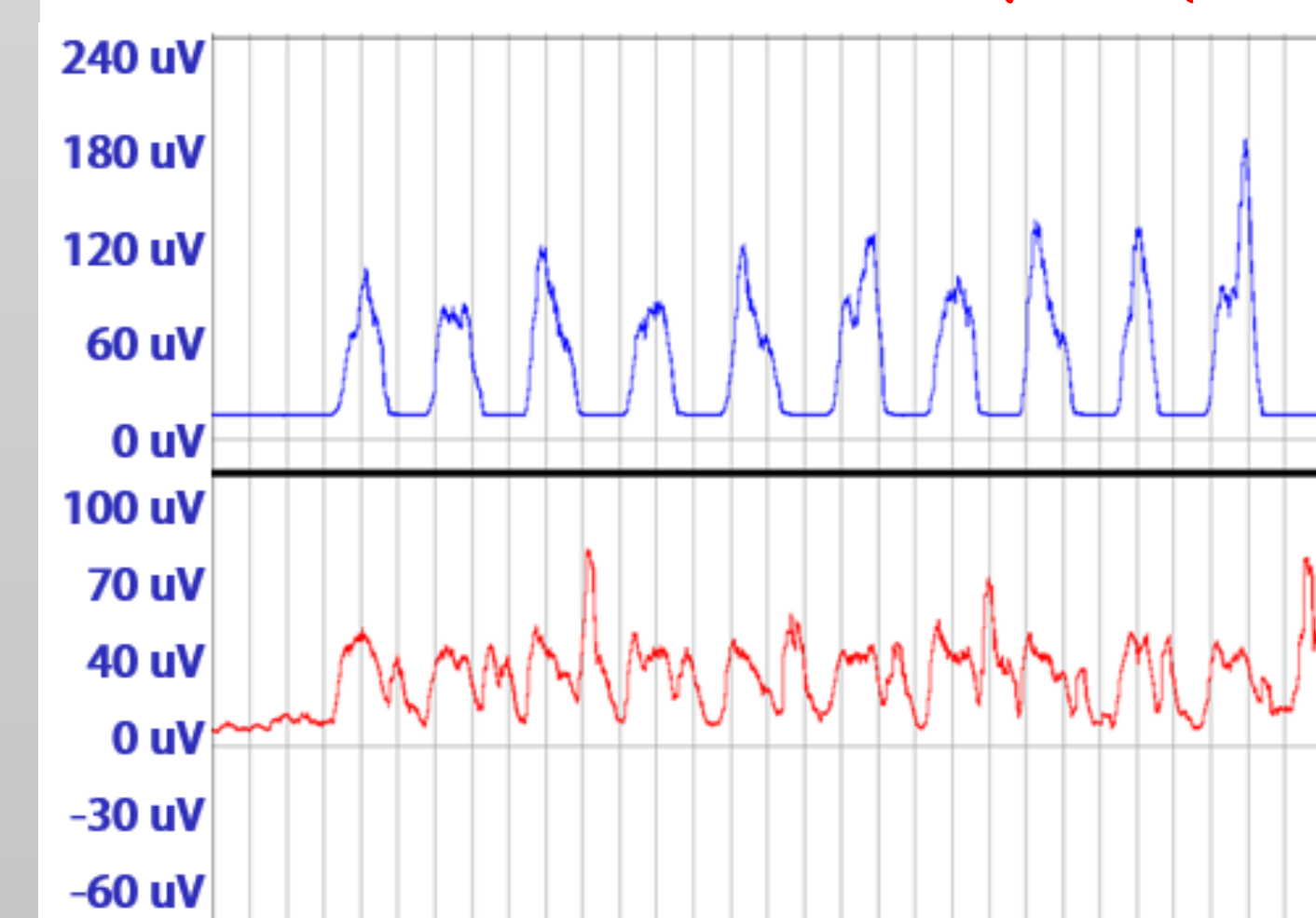
Target Muscles:
Tibialis (Shin)
and
Gastrocnemius (Calf)



Without Angled Seat + Footrest

Sensor 1) Gastroc. muscle activity = 120μV

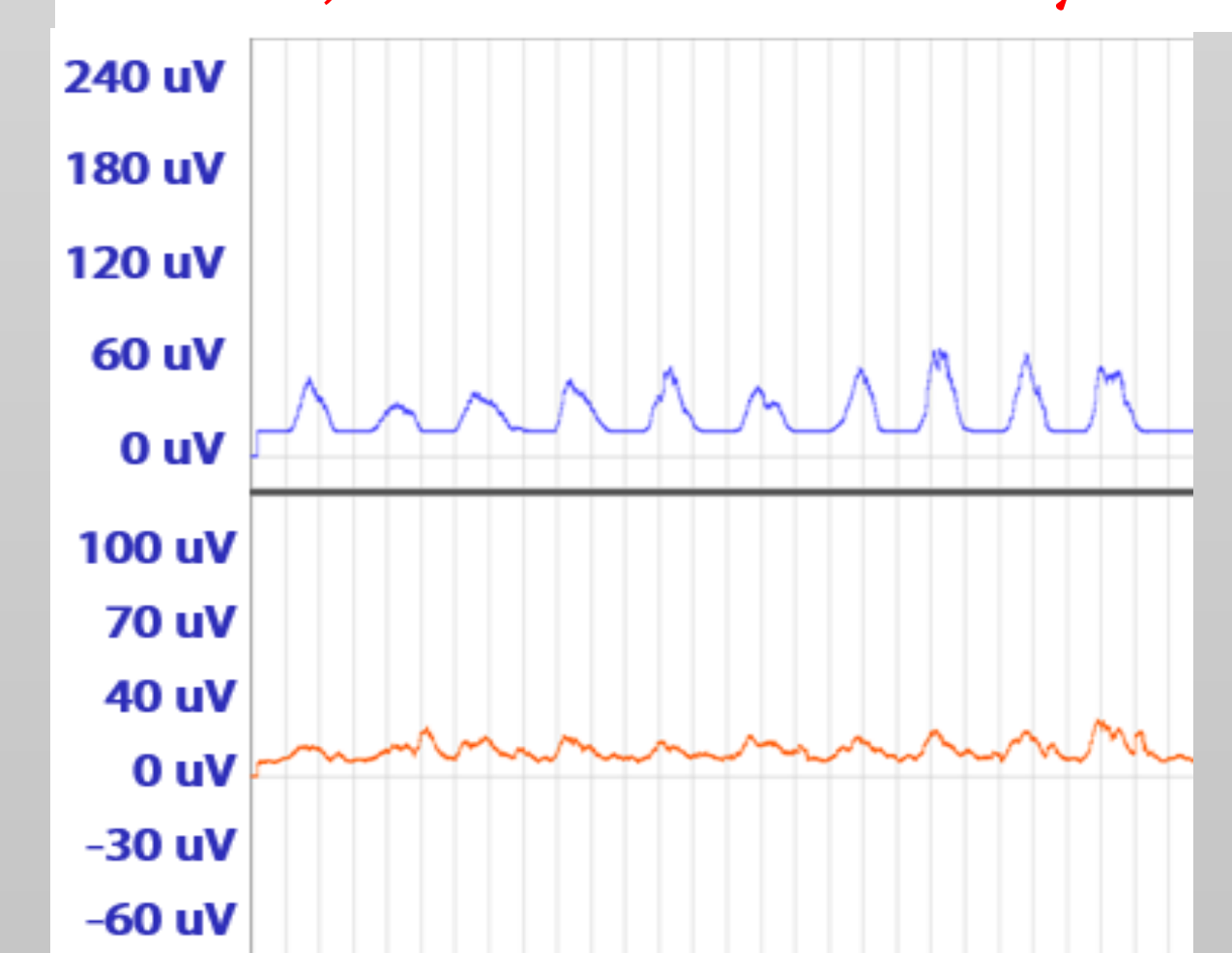
Sensor 2) Tibialis muscle activity = 48μV



With Angled Seat + Footrest

Sensor 1) Gastroc. muscle activity = 60μV

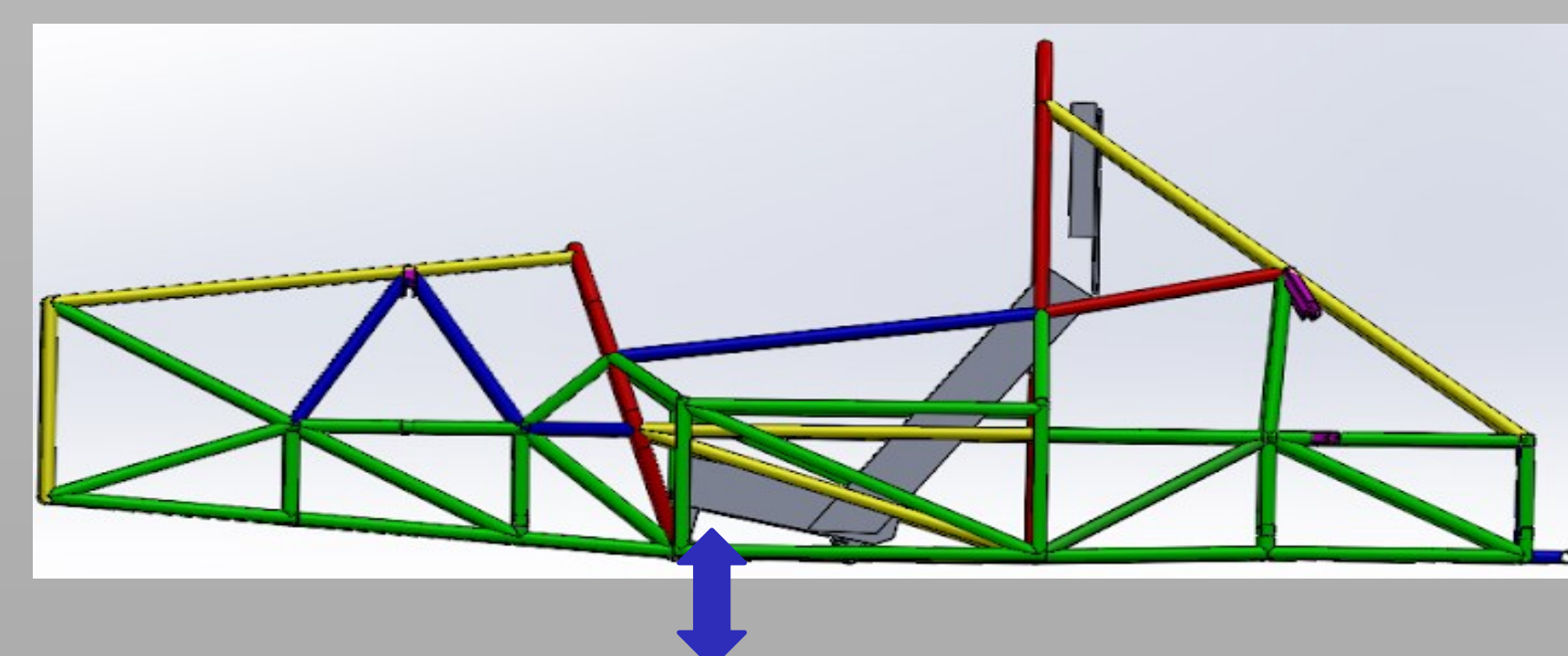
Sensor 2) Tibialis muscle activity = 12μV



This shows that with our modifications (Right), driver fatigue is greatly reduced.

Current Work

Operating the vehicle while using our pedal and seat designs results in less muscle activity in target muscles. The reduced muscle activity results in a greater number of cycles until fatigue onsets, meaning the driver will remain more comfortable for longer periods of time on the track.



To further optimize our seating system for drivers of all sizes, we plan on separating the back and bottom pieces of the seat. This will allow us to adjust the angle of the seat bottom to more comfortably fit very tall or very short drivers.

Acknowledgements

We would like to thank the 2017-2018 LTU Formula SAE Hybrid team for assisting us with our research and funding this project.